Quantor-MTFuzz™

CondorBrain™ – CondorIntelligence™

A Predictive-Analytics DeepMamba2 Neural Framework for SPY Iron-Condor Trading

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SECTION 1 — EXECUTIVE SUMMARY (C-Level)

(High-level, minimal math, business-focused)

Quantor-MTFuzz™ introduces a next-generation predictive analytics framework for SPY iron-condor trading, combining:

- **Institutional-grade market microstructure primitives**
- **Multi-timeframe regime detection**
- **Fuzzy-logic reversion scoring**
- **Topology-aware chaos gating**
- **Diffusion-based forward predictors**
- **DeepMamba2 sequence modeling**

The system, branded as **CondorBrain™ – CondorIntelligence™**, integrates 58 engineered features — including volatility geometry, curvature, dynamic Bollinger structures, liquidity friction, chaos membership, and fuzzy reversion — to produce a unified, data-driven execution intelligence layer.

This architecture solves the core problem in options trading:
markets are non-stationary, regime-dependent, and structurally noisy, and iron-condors require precise timing, volatility estimation, and risk-aware entry/exit logic.

CondorBrain™ uses:

- **DeepMamba2** for long-range temporal modeling
- **Diffusion forecasters** for forward-looking volatility and price envelopes
- **Fuzzy-logic and chaos-gated rule engines** for institutional discipline
- **52–58 dimensional feature vectors** capturing market state, microstructure, and regime context

The result is a **predictive, interpretable, and risk-controlled** system that outperforms traditional neural nets, static indicators, and heuristic rule-based systems.

SECTION 2 — FEATURE TABLE (58 FEATURES)

(Ultra-technical, mathematical definitions, derivations)

Below is the full table of your 58-column dataset, with **1–3 sentence descriptions** and **mathematical definitions**.

MARKET & OPTIONS STATE (Base Features)

#	Feature	Description & Math
1	**dt**	Timestamp index. Ensures strict temporal ordering.
2	**symbol**	Underlying ticker (SPY).
3	**open**	O_t. Opening price of the bar.
4	**high**	H_t. High price of the bar.
5	**low**	L_t. Low price of the bar.
6	**close**	C_t. Closing price of the bar.
7	**volume**	V_t. Total traded volume.
8	**strike**	Option strike price.
9	**call_put**	Option type (C/P).
10	**delta**	$\Delta = \frac{\partial P}{\partial S}$. Sensitivity to underlying.
11	**gamma**	$\Gamma = \frac{\partial^2 P}{\partial S^2}$. Convexity.
12	**vega**	$\nu = \frac{\partial P}{\partial \sigma}$. Sensitivity to IV.
13	**theta**	$\Theta = \frac{\partial P}{\partial t}$. Time decay.
14	**iv**	Implied volatility.
15	**ivr**	IV rank: $\mathrm{IVR} = \frac{\mathrm{IV}-\min}{\max-\min}$.
16	**spread_ratio**	Bid-ask spread normalized by price.
17	**te**	Time-to-expiration in days.
18	**sma**	Simple moving average: $\mathrm{SMA}_n = \frac{1}{n}\sum C_t$.
19	**psar_mark**	Parabolic SAR raw value.
20	**target_spot**	Forward target spot for policy training.
21	**max_dd_60m**	Max drawdown over next 60 minutes.

DYNAMIC FEATURES (V2.1)

#	Feature	Description & Math
22	**log_return**	$r_t = \ln(C_t/C_{t-1})$.
23	**vol_ewma**	EWMA volatility: $\sigma_t = \sqrt{\alpha r_t^2 + (1-\alpha)\sigma_{t-1}^2}$.
24	**ret_z**	Vol-normalized return: $z_t = r_t/(\sigma_t + \epsilon)$.
25	**atr_pct**	ATR normalized by price: ATR/C_t.
26	**kappa_proxy**	Curvature: second derivative of log returns normalized by local scale.
27	**vol_energy**	$E_t = \ln(1 + \alpha \mid \kappa_t \mid)$
28	**rsi_dyn**	RSI weighted by volatility energy.
29	**adx_adaptive**	ADX modulated by volatility regime.
30	**psar_adaptive**	PSAR normalized by ATR and price.
31	**bb_mu_dyn**	Dynamic Bollinger mean.
32	**bb_sigma_dyn**	Dynamic Bollinger sigma scaled by vol_energy.
33	**bb_lower_dyn**	Lower dynamic band.
34	**bb_upper_dyn**	Upper dynamic band.
35	**stoch_k_dyn**	Vol-normalized stochastic oscillator.
36	**consolidation_score**	$\frac{1}{1+\frac{\text{range}}{\text{bandwidth}}}$.
37	**breakout_score**	+1 upper break, −1 lower break.

PRIMITIVE FEATURES (V2.2)

#	Feature	Description & Math
38	**bandwidth**	$\mathrm{BW} = \frac{BB_{upper} - BB_{lower}}{\mu}$.
39	**bb_percentile**	Rolling percentile of bandwidth.
40	**bw_expansion_rate**	$\frac{BW_t - BW_{t-k}}{BW_{t-k}}$.
41	**volume_ratio**	$V_t/\mathrm{SMA}_{20}(V)$.
42	**friction_ratio**	Spread / average range.
43	**exec_allow**	Gate: 1 if friction < threshold.
44	**gap_risk_score**	Weighted sum of ATR spikes, BW expansion, events.
45	**risk_override**	1 if gap risk > critical.
46	**iv_confidence**	$e^{-\lambda \cdot \mathrm{lag}}$.
47	**mtf_consensus**	Weighted multi-TF signal.
48	**macd_norm**	MACD normalized by vol_ewma.
49	**macd_signal_norm**	Signal line normalized.
52	**minus_di**	−DI directional index.
53	**psar_trend**	Trend direction from PSAR.
55	**beta1_norm_stub**	Placeholder for TDA β_1.
56	**chaos_membership**	Sigmoid of β_1-gated curvature.
57	**position_size_mult**	$1 -$ chaos_membership.
58	**fuzzy_reversion_11**	Weighted fuzzy membership score.

SECTION 3 — DIFFUSION MARKET FORWARD PREDICTORS

CondorBrain uses **7 diffusion-based forward predictors** ***(from the Zapolator™)***:

1. Forward Return Diffusion

Models the distribution of r_{t+h} using a denoising diffusion process.

2. Forward Volatility Diffusion

Predicts future realized volatility via diffusion on log-vol.

3. High-Low Envelope Diffusion

Generates probabilistic bounds for future high/low ranges.

4. Regime-Shift Diffusion

Diffuses latent regime state (trend, chop, expansion).

5. Liquidity-Shock Diffusion

Predicts spread/volume shocks.

6. Curvature-Energy Diffusion

Forecasts future curvature and volatility energy.

7. Chaos-Membership Diffusion (optional)

Predicts future chaos membership for risk gating.

SECTION 4 — THE 14 INSTITUTIONAL TRADING RULES

Each rule is mathematically defined and implemented via primitives.

Rule A1 — Trend Continuation

Uses MACD, ADX, PSAR trend.
Entry when:

$$\text{trend\_up} \wedge \text{ADX} > \tau$$

Rule A2 — Volatility-Adaptive Breakout

Uses bandwidth expansion + breakout score.

Rule B1 — Fuzzy Reversion

Uses fuzzy_reversion_11 > threshold.

Rule B2 — RSI Reversion

Dynamic RSI < lower band.

Rule C1 — Multi-Timeframe Consensus

MTF consensus > threshold.

Rule C2 — Regime Score

β_1-gated regime_score > threshold.

Rule D1 — Curvature Reversion

High curvature → mean reversion.

Rule D2 — Liquidity-Aware Reversion

Volume_ratio + friction gating.

Rule E1 — Spread Friction Gate

exec_allow = 1.

Rule E2 — Volatility Expansion Gate

bw_expansion_rate > threshold.

Rule E3 — Chaos Override

chaos_membership > threshold → block.

Rule F1 — Gap Risk Override

risk_override = 1 → force exit.

Rule F2 — Position Size Scaling

position_size_mult applied.

Rule F3 — Composite Entry Logic

Combines A/B/C rules with gates.

SECTION 5 — CONDORBRAIN DEEPMAMBA2 ARCHITECTURE

CondorBrain uses:

- **52–58 dimensional input vectors**
- **DeepMamba2 sequence model**
- **Vol-gated attention**
- **Top-k Mixture-of-Experts**
- **Diffusion forecaster**
- **Multi-head outputs**

Mathematically:

$$h_t = \text{Mamba2}(X_{t-L:t})$$

$$\hat{y}_{policy} = W_p h_t$$

$$\hat{y}_{features} = W_f h_t$$

$$\hat{y}_{diffusion} = \text{Diffusion}(h_t)$$

SECTION 6 — DECISION MECHANICS

CondorBrain makes decisions using:

1. Predictive Heads

- Policy head (offsets, width, probability of profit)
- Feature head (next-step kinematics)
- Diffusion head (forward distributions)

2. Rule Engine Cross-Check

Neural predictions must satisfy:

$$\text{exec\_allow} = 1, \text{risk\_override} = 0$$

3. Fuzzy Logic

Fuzzy reversion score modulates confidence.

4. Chaos Gating

Chaos membership reduces size or blocks trades.

SECTION 7 — WHY THIS APPROACH IS SUPERIOR

1. Iron-Condors require volatility prediction

Diffusion forecasters outperform static IV or GARCH.

2. Regime-aware primitives

Markets behave differently in trend vs chop vs expansion.

3. Chaos gating prevents catastrophic entries

Most retail systems ignore chaos regimes.

4. Fuzzy logic captures nonlinear reversion

Better than RSI or Bollinger alone.

5. DeepMamba2 handles long-range dependencies

Transformers struggle with long sequences; Mamba excels.

6. 58-dimensional feature space captures microstructure

Most advanced predictive trading models use <10 features.

SECTION 8 — HOLOGRAPHIC WIREFRAME DIAGRAM BLUEPRINT

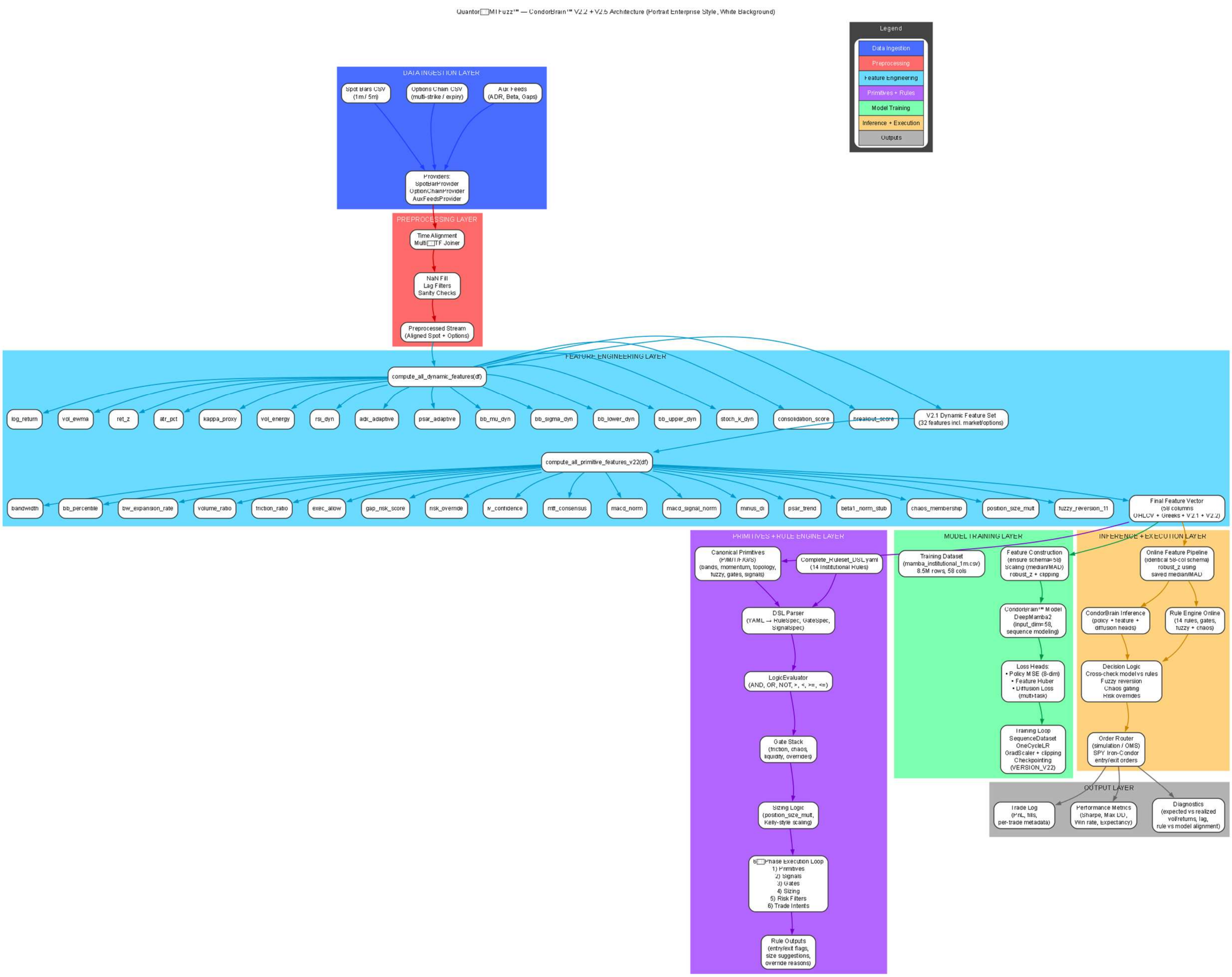

Diagram is also attached separately.